

Run 09 | NME

Global (including India)

IV + controls

Target Necessity-motivated entrepreneurship; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 69,931 (5.21%) **Case time** 62m 17s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9796). Highest F1: Random Forest (0.6642). For XGBoost, adding controls changed ROC-AUC by +0.0033 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9796	0.7154	96.22	62.03	70.76	0.6611	24s
CatBoost	0.9793	0.7125	96.20	61.80	70.91	0.6605	2m 58s
Random Forest	0.9789	0.7160	96.26	62.42	70.96	0.6642	2m 59s
H2O GBM	0.9778	0.6911	96.02	60.32	69.03	0.6438	1m 00s
Elastic Net	0.9752	0.6608	95.81	58.50	67.31	0.6260	16m 31s
Lasso	0.9752	0.6608	95.81	58.52	67.28	0.6260	20m 13s
Logistic Regression	0.9750	0.6591	95.80	58.42	67.15	0.6248	2m 31s
Ridge	0.9738	0.6421	95.37	54.15	72.30	0.6192	9m 44s
Decision Tree	0.9688	0.5880	96.09	62.93	60.41	0.6164	13s
Naive Bayes	0.8851	0.2272	89.06	25.72	58.32	0.3569	4m 50s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	16,586.69	29.19
2	Established business owner	7,384.25	12.99
3	Occupational status	3,849.34	6.77
4	Survey year	2,666.69	4.69
5	Age	2,518.45	4.43
6	Household size	1,449.00	2.55
7	Education	1,125.62	1.98
8	Household income	1,054.82	1.86

Share is the percentage of total forest impurity importance across all 66 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Education	1,125.62	1.98
2	Living conditions	986.67	1.74
3	Safety and security	839.66	1.48

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 09_obj2_global_iv_cv_NME. Scores and importance are taken directly from saved CSV outputs.